

NEMA Intelligent Lighting Controller

LoRaWAN | 4G

A sustainable solution for smart cities of tomorrow



NEMA Intelligent Lighting Controller – LoRaWAN & 4G

Vialight's NEMA ILC is a vital part of smart lighting solutions. It manages street light ON/OFF schedules, monitors temperature, and offers over and under voltage protection. It also measures power, energy, current, and voltage, helping track and optimize overall energy consumption.



Features



Robust and Reliable

Offer long range, deep coverage, high availability and managed Low Power Wide Area Network (LPWAN).



Remote Controlling

Supports multiple control modes, including Manual and Configurable ON/OFF/Dimming.



Power Measurements

High performance Analog front end solutions measuring electrical parameters such as Current, Voltage, Frequency, Power Factor, kW and kWh.



Grouping

Smart grouping of lights based on application needs, enabling efficient remote group-wise control.



Dimming Controls

Supports dimming via 0-10 VDC, PWM with up to 10 dimming schedules.



Fault Monitoring

Reports on lamp burnouts, abnormal power consumptions, low power factor, and more via cloud.



Alerts

Notifies users via e-mail or text (SMS) messages as soon as it occurs.

Technical Specification – LoRaWAN

Hardware	
Processor	Cortex-M4 STM32WL 32bit, RISC 48Mhz
Antenna Options	Internal Antenna
Mounting Type	NEMA 7 Pin Standard, Transparent - Grey
IP Enclosure Rating	IP66
Metering Measurement	Voltage, Current, Frequency, Power Factor
Real Time Clock	Battery - Backup RTC
Networking and Security	
Protocol	LoRaWAN
Frequency Band	865 to 867 MHz, 902 to 915 MHz
Encryption	AES 128 Bit for MAC and Application (DTLS) layers
Open Field Range	Upto *10000 meters (LOS)
Data Rate	37.5 Kbps Max
TX Power	+20 dBm
RX Sensitivity	-135 dBm (SF12)
Software and Control	
Notification	Lamp Over Voltage, Lamp Under Voltage, Light Fault, Meter Chip Fault, and Relay Fault
Monitoring Parameter	Voltage, Current, Power factor, Frequency, Power (kW), Energy (kWh)
Central Management System	Web-based software that allows remote configuration, scheduling, monitoring control and reporting
Electrical Specification	
Supply Type	Single Phase
Power Supply Range	90 VAC to 528 VAC, 50/60 Hz
Operating Temperature	-10°C to +60°C
Humidity Range	10% to 90% RH
Surge Protection	IEC/EN61000-4-5, 10KV
Switching Load Capacity	10A, 230 VAC
High Voltage Cutoff	290 VAC
Certificate	
Compliance	IEC/EN61000-4-5
IP Enclosure Ratings	IP66
LoRaWAN Module	WPC

Technical Specification – 4G

Hardware	
Processor	A7672S-FASE-SIMCOM
Antenna Options	Internal Flexible PCB Antenna
Mounting Type	NEMA 7 Pin Standard, Transparent - Grey
IP Enclosure Rating	IP66
Metering Measurement	Voltage, Current, Frequency, Power Factor
Real Time Clock	Battery - Backup RTC
Ambient Sensor Support	Detect amount of light in vicinity and based on that turn ON/OFF the light
Networking and Security	
Protocol	GSM / GPRS (4G /LTE) with 2G fallback
Diagnostic	Self diagnostic check
Latency	Operation response time: < 5 sec Data update time on LMS < 10 min
Data update frequency to server	Every 15 min (configurable up to 1 min)
Software and Control	
Notification	Lamp Over Voltage, Lamp Under Voltage, Light Fault, Meter Chip Fault, and Relay Fault
Monitoring Parameter	Voltage, Current, Power factor, Frequency, Power (kW), Energy (kWh)
Central Management System	Web-based software that allows remote configuration, scheduling, monitoring control and reporting
Electrical Specification	
Supply Type	Single Phase
Power Supply Range	176VAC to 528VAC, 50/60 Hz
Operating Temperature	-10°C to +60°C
Humidity Range	10% to 90% RH
Surge Protection	IEC/EN61000-4-5, 10KV
Switching Load Capacity	10A, 230 VAC
High Voltage Cutoff	290 VAC
Certificate	
Compliance	IEC/EN61000-4-5
IP Enclosure Ratings	IP66

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